Countable Capture for Von Neumann Algebras

Run `fa_banach_001`, lane 19

June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. It proves that a von Neumann algebra has BCP, equivalently UBCP, if and only if it is atomic and can be represented on a separable Hilbert space. The same paper asks for a broader noncommutative C^*-algebra version:

> $\{U_\alpha\}_{\alpha \in A}$ of K is called a π-basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].
>
> In the setting of commutative C^*-algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.
>
> **Corollary 3.9.** *Let $\mathcal{A}$ be a unital commutative C^*-algebra. Then the followings are equivalent:*
>
> (1) *$\mathcal{A}$ has the BCP;*
> (2) *the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π-basis;*
> (3) *$\mathcal{A}$ has the UBCP.*
>
> In particular, there exist commutative C^*-algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].
>
> We conclude this section by the following question.
>
> **Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^*-algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Theorem 1. *For a von Neumann algebra M, the following are equivalent:*

1. *M has the countable-capture property;*
2. *M fails BCP;*
3. *M is not atomic and separably represented.*

Here countable capture is understood in the Banach-space sense: for every countable $C \subset M$, there is $u \in S_M$ such that

$$\operatorname{dist}(c, \mathbb{C}u) = \|c\| \qquad (c \in C).$$

This is equivalent to the contraction formulation in the packet

`2311.14257_countable_capture_characterization`.

Idea of the Proof

Countable capture always implies failure of BCP. The source paper gives the von Neumann BCP classification, so it remains to prove that every von Neumann algebra outside the atomic separably represented class has countable capture.

There are two mechanisms. In the atomless part, the source singular-state lemma gives, for each centre x_n, a singular state φ_n with

$$\varphi_n(x_n^* x_n) = \|x_n\|^2.$$

A standard projection characterization of singular functionals lets us find one nonzero projection p with $\varphi_n(p) = 0$ for all n. Then the functionals

$$y \mapsto \frac{\varphi_n(x_n^* y)}{\|x_n\|}$$

norm x_n and all vanish at p, giving capture by the common-kernel form of the characterization.

In the atomic nonseparable case, either there are uncountably many atomic summands, which gives primitive-topological capture, or there is a summand $B(H)$ with H nonseparable. A countable family of operators on H has a separable reducing subspace that norms all of them; a projection orthogonal to that subspace gives the common capture direction.

Singular States and Common Kernels

We use the following standard characterization of singular positive functionals on von Neumann algebras.

Lemma 1 (common projection for countably many singular states)**.** *Let N be a nonzero von Neumann algebra and let (φ_n) be countably many singular states on N. Then there is a nonzero projection $p \in N$ such that*

$$\varphi_n(p) = 0 \qquad (n \geq 1).$$

Proof. Fix a normal state ψ on N. A singular positive functional φ admits an increasing net of projections $e_i \uparrow 1$ with $\varphi(e_i) = 0$. Hence, by normality of ψ, for every $\varepsilon > 0$ there is a projection e such that

$$\varphi(e) = 0, \qquad \psi(1 - e) < \varepsilon.$$

Apply this to φ_n with $\varepsilon_n = 2^{-n-2}$, obtaining projections e_n with $\varphi_n(e_n) = 0$ and $\psi(1-e_n) < 2^{-n-2}$. Put

$$p = \bigwedge_{n=1}^{\infty} e_n.$$

Then

$$\psi(1-p) = \psi\Big(\bigvee_n (1 - e_n)\Big) \leq \sum_n \psi(1 - e_n) < \frac{1}{2}.$$

Thus $p \neq 0$. Since $p \leq e_n$, we have $\varphi_n(p) \leq \varphi_n(e_n) = 0$ for all n. □

Lemma 2 (atomless summands have capture)**.** *Let N be an atomless von Neumann algebra. Then N has countable capture.*

Proof. Let $C = \{x_1, x_2, \ldots\} \subset N$ be countable. Discard the zero elements. By the singular-state norming lemma from the source paper, for each $x_n \neq 0$ there is a singular state φ_n on N such that

$$\varphi_n(x_n^* x_n) = \|x_n\|^2.$$

By Lemma 1, choose a nonzero projection $p \in N$ with $\varphi_n(p) = 0$ for every n. Replacing p by its support projection if necessary is unnecessary; already $\|p\| = 1$.

For each n, define

$$f_n(y) = \frac{\varphi_n(x_n^* y)}{\|x_n\|}, \qquad y \in N.$$

By Cauchy–Schwarz for the positive functional φ_n,

$$|f_n(y)| \leq \frac{\varphi_n(x_n^* x_n)^{1/2} \varphi_n(y^* y)^{1/2}}{\|x_n\|} \leq \|y\|.$$

Moreover

$$f_n(x_n) = \|x_n\|,$$

so f_n is a norming functional for x_n. Finally,

$$|f_n(p)|^2 = \frac{|\varphi_n(x_n^* p)|^2}{\|x_n\|^2} \leq \frac{\varphi_n(x_n^* x_n)\varphi_n(p)}{\|x_n\|^2} = 0.$$

Thus all f_n's vanish at the same nonzero vector p. The common-kernel form of countable capture applies. □

Atomic Nonseparable Cases

Lemma 3. *If H is a nonseparable Hilbert space, then $B(H)$ has countable capture.*

Proof. Let $C = \{T_1, T_2, \ldots\} \subset B(H)$ be countable. Choose a separable closed subspace $L \subset H$ which reduces every T_n and satisfies

$$\|T_n|_L\| = \|T_n\| \qquad (n \geq 1).$$

This is obtained by starting with countably many almost norming vectors for the T_n's and closing under all T_n and T_n^*. Since H is nonseparable, $L^\perp \neq 0$. Let p be a nonzero projection with range contained in $L^\perp$.

For every scalar λ,

$$(T_n - \lambda p)|_L = T_n|_L.$$

Hence

$$\|T_n - \lambda p\| \geq \|T_n|_L\| = \|T_n\|.$$

Taking the infimum over λ gives

$$\operatorname{dist}(T_n, \mathbb{C}p) = \|T_n\|,$$

so $B(H)$ has countable capture. □

Lemma 4. *Let M be an atomic von Neumann algebra which is not separably represented. Then M has countable capture.*

Proof. Write
$$M \cong \prod_{i\in I} B(H_i)$$
as an atomic von Neumann algebra.

If I is uncountable, then the coordinate ideals give an uncountable family of pairwise orthogonal nonzero closed two-sided ideals. Hence $\mathrm{Prim}(M)$ has no countable π-base, and the primitive-space countable-capture theorem from the packet

`2311.14257_countable_capture_characterization`

implies countable capture.

If I is countable, nonseparable representability means that at least one summand $B(H_{i_0})$ has H_{i_0} nonseparable. Let
$$C = \{x_1, x_2, \ldots\} \subset M$$
be countable, and write $x_n = (x_{n,i})_{i\in I}$. Apply Lemma 3 in $B(H_{i_0})$ to the countable family (x_{n,i_0}), obtaining a projection $p \in B(H_{i_0})$ such that
$$\mathrm{dist}(x_{n,i_0}, \mathbb{C}p) = \|x_{n,i_0}\| \qquad (n \geq 1).$$
Let $u \in M$ be p in the i_0-coordinate and 0 elsewhere. Then $\|u\| = 1$. For every scalar λ,
$$\|x_n - \lambda u\| = \max\left\{ \sup_{i\neq i_0} \|x_{n,i}\|, \|x_{n,i_0} - \lambda p\| \right\} \geq \max\left\{ \sup_{i\neq i_0} \|x_{n,i}\|, \|x_{n,i_0}\| \right\} = \|x_n\|.$$
Therefore $\mathrm{dist}(x_n, \mathbb{C}u) = \|x_n\|$ for all n, giving countable capture. □

Proof of the Classification

Proof of Theorem 1. If M has countable capture, then M fails BCP by the general countable-capture obstruction packet.

The source paper proves that a von Neumann algebra has BCP if and only if it is atomic and separably represented. Thus BCP failure is equivalent to being outside that class.

It remains to show that every von Neumann algebra outside that class has countable capture. Let z be the central projection supporting the atomless part of M. If $z \neq 0$, then zM has countable capture by Lemma 2. Given a countable $C \subset M$, apply that lemma to zC and obtain $u \in zM$ with $\|u\| = 1$ and
$$\mathrm{dist}(zc, \mathbb{C}u) = \|zc\| \qquad (c \in C).$$
Since u is zero on $(1-z)M$, for every scalar λ,
$$\|c - \lambda u\| = \max\{\|(1-z)c\|,\ \|zc - \lambda u\|\} \geq \max\{\|(1-z)c\|,\ \|zc\|\} = \|c\|.$$
Hence M has countable capture.

If $z = 0$, then M is atomic. Since M is outside the atomic separably represented class, it is atomic but not separably represented. Lemma 4 gives countable capture. □

Scope and Use for the C*-Problem

This gives a complete countable-capture classification for von Neumann algebras:

$$\text{BCP failure} \quad \Longleftrightarrow \quad \text{countable capture.}$$

It supports the broader program of finding whether C*-algebraic BCP failure is always witnessed by countable capture, while also showing the different mechanisms needed: primitive topology, nonseparable atomic Hilbert summands, and singular-state capture in atomless summands.

Novelty and Search Bounds

The source paper already proves the von Neumann BCP/UBCP classification and the singular-state norming lemma for atomless von Neumann algebras. The local indexes were searched for countable capture, von Neumann algebras, singular states, atomless, atomic, and separably represented. Web searches on June 17, 2026 for BCP and von Neumann classification found the source paper but no separate countable-capture reformulation.

Human Review Focus

Reviewers should check:

1. Lemma 1, especially the standard singular functional projection characterization;
2. the use of the source singular-state norming lemma in Lemma 2;
3. the atomic nonseparable decomposition in Lemma 4;
4. the final central-summand maximum-norm argument.

References

[1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.

[2] Masamichi Takesaki, *Theory of Operator Algebras I*, Springer, 1979.

[3] Run `fa_banach_001`, lane 19, *When the Countable-Capture BCP Obstruction Exists*, solution packet `2311.14257_countable_capture_characterization`, 2026.

[4] Run `fa_banach_001`, lane 19, *A Primitive-Ideal Pi-Base Obstruction for C*-Algebras with BCP*, solution packet `2311.14257_primitive_pi_base_obstruction`, 2026.